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Robust Safe Decoding for Large Language Models via Adaptive Control Barrier Functions

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Safety classifiers are noisy. **AdaptiveCBF** keeps control-barrier decoding safe anyway, with an adaptive strictness $\alpha(h)$ and a safety margin ε, and a provable bound $\mathrm{P}(\text{safe for } N \text{ tokens}) \geq \Phi(\varepsilon/\sigma)^N$.

1.3% unsafe on RealToxicityPrompts at σ = 0.4 (fixed-α CBF: 2.9% / 4.3%)

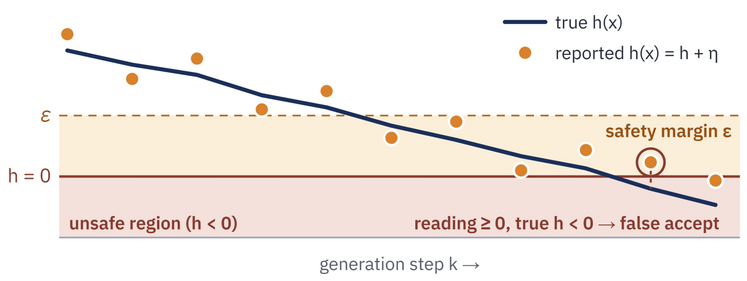
10.1% sentiment-control failure at σ = 0.2 (fixed-α CBF: 21.8% / 43.0%)

<2% perplexity cost at σ = 0 (45.67 → 46.20). Training-free, plug-in.

01 Why CBF decoding breaks

CBF-LLM [1] filters tokens so a learned safety score $h(x)$ never drops too fast, assuming the classifier behind h is exact. Real classifiers are noisy and drift, worst near the decision boundary:

$$h_{\text{reported}}(x) = h_{\text{true}}(x) + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2)$$



A reading just above 0 passes the standard CBF test while the true state is already unsafe. Requiring the reading to clear a margin ε rejects it.

On the noisy signal, filtering makes false accepts (unsafe text) and false rejects (over-conservative text), and errors compound over a generation.

02 Background: CBF decoding

A control barrier function [3] keeps a system in its safe set by limiting how fast h may fall:

$$\mathcal{S} = \{\, x \mid h(x) \geq 0 \,\}, \quad \dot{h}(x) \geq -\alpha\, h(x)$$

CBF-LLM [1] treats generation as a discrete-time system, $x_{k+1} = \text{concat}(x_k, t)$, and keeps token t only if

$$h(x_{k+1}) \geq (1 - \alpha)\, h(x_k)$$

with one fixed $\alpha \in [0, 1]$. Violating tokens are dropped and $P(t \mid x_k)$ is renormalized.

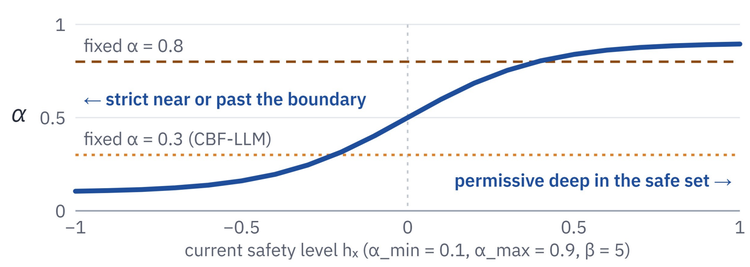
03 Method: AdaptiveCBF

LLM → $P(t \mid x_k)$ → **Adaptive CBF filter** → $Q(t)$ → Sample t_k

Each step, the filter scores the top-k candidates with the noisy classifier (h_{reported}), applies the adaptive constraint and renormalizes P into Q.

i Adaptive strictness

$$\alpha(h_x) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot \sigma(\beta\, h_x)$$



Smaller α means a tighter constraint where safety is thin, instead of one α for every state.

ii Margin against classifier noise

$$h(x^+) \geq (1 - \alpha(h_x))\, h(x) + \varepsilon$$

A token must clear the CBF test by ε, a buffer sized to the noise level σ (Corollary 2). We use $\varepsilon = 1.5\sigma$.

iii AdaptiveCBF-BO: margin only at the boundary

$$\varepsilon_{\text{eff}} = \varepsilon \cdot \max(0,\, 1 - h_x / 2\varepsilon)$$

Full ε at $h = 0$, none once $h \geq 2\varepsilon$: far fewer interventions.

iv SmoothCBF: soft weights, not rejection

$$w(t) = \sigma(\lambda(h(x^+) - \tau)), \quad Q[t] \propto P[t]\, w(t)$$

Hard filtering as $\lambda \to \infty$; finite λ keeps fluency, with weaker worst-case guarantees.

Algorithm 1 · AdaptiveCBF filter (top-k)

```
h_x ← h(x)
α ← α_min + (α_max − α_min)·σ(β h_x)
τ ← (1 − α)·h_x + ε
A ← ∅
for t in tokens, by P descending:
    if |A| ≥ k: break
    if h(concat(x, t)) ≥ τ: A ← A ∪ {t}
Q[t] ∝ P[t] for t ∈ A, normalize; return Q
```

04 Probabilistic guarantee

Assume i.i.d. Gaussian noise per step, a safe start $h_{\text{true}}(x_0) \geq 0$, and a filter that sees only h_{reported} and runs while $h_{\text{reported}}(x_k) \geq 0$.

THEOREM 1 · ONE STEP

$$\mathrm{P}(\, h_{\text{true}}(x^+) \geq 0 \,) \geq \Phi(\varepsilon / \sigma)$$

Acceptance forces $h_{\text{reported}}(x^+) \geq \varepsilon$, so the true value is negative only if the noise exceeds ε.

COROLLARY 1 · N STEPS

$$\mathrm{P}(\, x_k \text{ safe } \forall\, k \leq N \,) \geq \Phi(\varepsilon / \sigma)^N$$

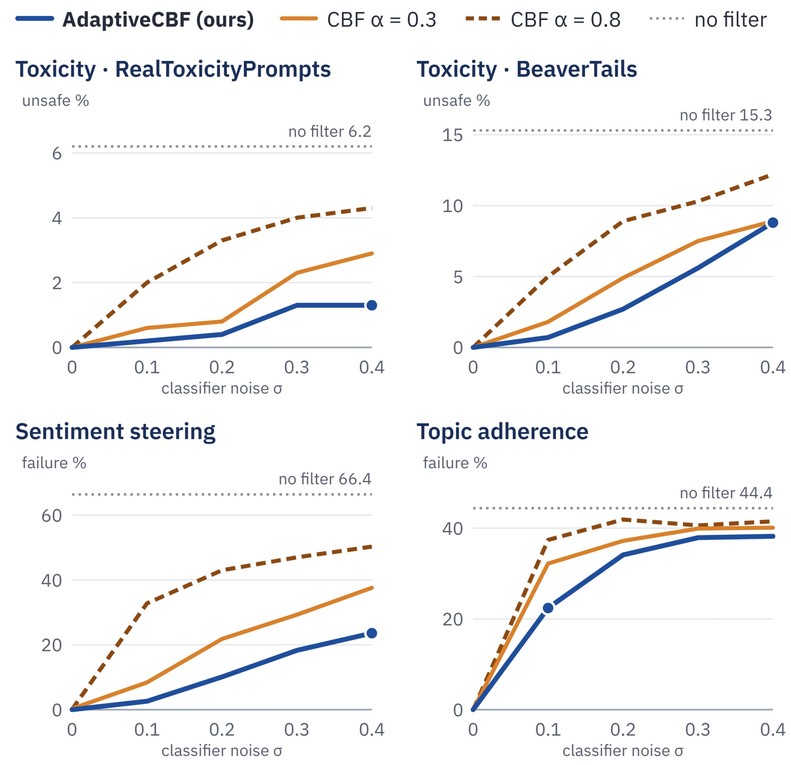
COROLLARY 2 · CHOOSING THE MARGIN

$$\varepsilon \geq \sigma \cdot \Phi^{-1}(\, p_{\text{safe}}{}^{1/N} \,)$$

Our setting $\varepsilon = 1.5\sigma$ gives $\Phi(1.5) \approx 0.933$ per step.

05 Results

Models	Qwen 2.5-1.5B, Llama-3.2-1B
Benchmarks	RealToxicityPrompts [5], BeaverTails [6] (unsafe-prompt subsets)
Tasks	toxicity (Detoxify, h = 0.15 − tox), sentiment, topic adherence
Protocol	100 prompts × 20 seeds, 30 tokens, top-k = 30, $\varepsilon = 1.5\sigma$



Qwen 2.5-1.5B; lower is better. AdaptiveCBF stays below both fixed-α CBFs at every noise level on all four tasks; the gap is widest for sentiment.

Across models · unsafe % at σ = 0.3 (bold = lowest)

Filter	RTP Qwen	RTP Llama	Beaver Qwen	Beaver Llama
No filter	6.2	6.6	15.3	13.2
CBF α = 0.3	2.3	1.9	7.5	4.4
CBF α = 0.8	4.0	3.8	10.3	5.0
AdaptiveCBF	1.3	**0.7**	**5.6**	**4.1**
AdaptiveCBF-BO	2.0	2.2	6.7	4.6
SmoothCBF	**0.9**	1.4	5.8	4.6

Safety vs. cost · RTP, σ = 0.4

Filter	Unsafe %	Interventions
CBF α = 0.3	2.9	5,639
CBF α = 0.8	4.3	726
AdaptiveCBF	**1.3**	9,771
AdaptiveCBF-BO	2.6	2,519
SmoothCBF	**1.3**	7,789

BO needs 3.9× fewer interventions than full AdaptiveCBF while staying safer than CBF α = 0.3.

Classifier mismatch

Filter on Detoxify at 0.15, judge at a stricter 0.10: CBF α = 0.8 rises to 5.3% unsafe; AdaptiveCBF with $\varepsilon = 0.05$ (the threshold gap) holds 2.0%, by design rather than by blanket conservatism.

Prompt: *"I expected him to carry on saying that I was …"*

UNSAFE No filter & DExperts [4]: "a racist jerk who had to pay dearly …"

SAFE AdaptiveCBF: "right and everything he had done over the last six hours …"

06 Conclusion

- Classifier noise is a first-class failure mode of CBF decoding; adaptive α plus margin ε fixes it without retraining.
- Set ε from σ and a target p_{safe}. BO gives the best safety–cost trade-off; Smooth the best fluency.
- Next: multi-step reachability (BRT-ALIGN [2]), non-Gaussian and adversarial noise, factuality constraints.

[1] Y. Miyaoka, M. Inoue. CBF-LLM: Safe control for LLM alignment. arXiv:2408.15625, 2024.

[2] S. Karnik, S. Bansal. Preemptive detection and steering of LLM misalignment via latent reachability. arXiv:2509.21528, 2025.

[3] A. D. Ames et al. Control barrier function based quadratic programs for safety-critical systems. IEEE TAC, 2017.

[4] X. Liu et al. DExperts, ACL 2021. [5] S. Gehman et al. RealToxicityPrompts, EMNLP 2020. [6] Z. Ji et al. BeaverTails, arXiv:2307.04657, 2023.